Shaurya Aswal

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About Me

Computer Science student (2028) focused on data science and machine learning — specifically where models drive real business decisions. My work spans causal inference for customer retention, demand forecasting for supply chains, and front-end development in a startup. I'm drawn to problems like fraud detection, churn prediction, risk scoring, and revenue optimization — where a model's output actually changes a decision someone makes, not just sits in a notebook.

Education

Maharaja Surajmal Institute of Technology **Bachelor of Technology • 8.85 / 10.00**
Computer Science Engineering August 2024 – August 2028

Skills

ML & Data Science : Predictive Modeling, Uplift Modeling (Causal Inference), Time Series Forecasting, A/B Testing, Feature Engineering
Business Analytics : Credit Risk, Fraud Detection, Churn Prediction, Customer Lifetime Value, Revenue Optimization
Generative AI & LLMs : Prompt Engineering, Fine-Tuning, LangChain, NLP, Vision Models, MCP Servers
Languages & Libraries : Python, SQL, Pandas, NumPy, XGBoost, Scikit-Learn, TensorFlow, CausalML, Matplotlib, Seaborn
Tools & Databases : MongoDB, Git, OpenCV, Data Pipeline Design | Web: React, Node.js, JavaScript, HTML/CSS
MLOps : MLflow, DVC, Docker

Projects

Customer Retention & Revenue Uplift Model

Tech: Python, XGBoost, CausalML, Scikit-Learn, Pandas

- Modeled persuadable customers using XGBoost uplift model tuned on Qini coefficient — targeting only those where a discount genuinely shifts the outcome, not those who stay or churn regardless.
- Selective targeting strategy drove $3.4M in recovered revenue while reducing discount spend vs. a random targeting baseline.
- Validated via synthetic A/B test across 120,000+ hotel reservations; model improved Qini score by 38% over the random targeting benchmark.

Predictive Demand & Inventory Forecasting System

Tech: Python, Scikit-Learn, Time Series Analysis, Statistical Modeling

- Forecasted weekly sales across 45 Walmart stores, 81 departments, spanning 3 years of data — using lag features, rolling averages, and Fourier seasonal terms.
- Reduced forecast error (RMSE) by ~22% vs. moving average baseline on the holdout set using walk-forward validation to prevent data leakage.
- Translated forecasts into inventory decisions — flagging top 10 overstocked and understocked department-store pairs per week.

Plant Disease Detection using CNN

Tech: Python, TensorFlow, Keras, Flask, Next.js, React, TypeScript, Tailwind CSS

- Trained a CNN on the PlantVillage dataset for multi-class plant disease classification from leaf images with confidence scoring.
- Built Flask REST APIs for model inference and a Next.js/React frontend with camera capture, image upload, and treatment recommendations.

Experience

Front-End Developer — TourDesigner **Jan 2025 – Feb 2025**

- Built React UI components, fixed cross-device bugs, and refactored routing logic — improved performance via lazy loading and cleaner component structure.
- Shipped fast iterations in a small team with shifting requirements, aligning output with founder priorities over original scope.

Societies & Leadership

IEEE Society — Project Committee Member
Actively involved in organizing and executing technical initiatives within the MSIT chapter.

Google Developers Group (GDG) on Campus — Lead
Mentored 200+ students for a Google campaign; top 70 performers were awarded Google goodies for their results.